1.1.1 Why norms?

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Week 01 due Oct 18, 2023 20:13 IST

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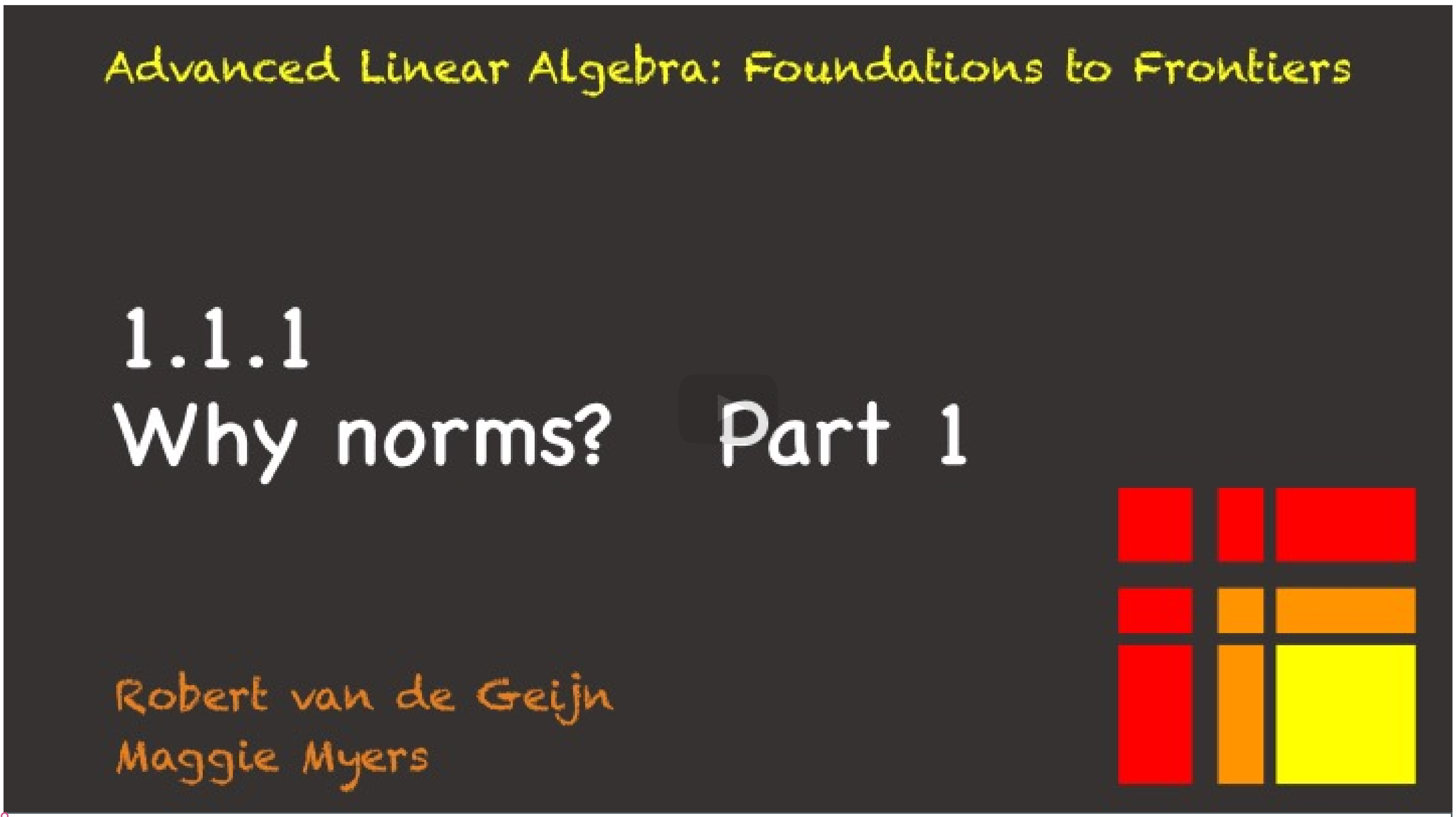

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Reading assignment

1/1 point (graded)

Read Unit 1.1.1 of the notes. [LINK]

- [x] done

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HW 1.1.1.3 - I Guess 0 is "small" 2

Homework 1.1.1.1.

1/1 point (graded)

Compute $\begin{pmatrix} 1 & -2 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} =$

-4 Answer: -4

-3 Answer: -3

2 Answer: 2

Solution:

For a complete solution see the notes .

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Answers are displayed within the problem

Homework 1.1.1.2

1/1 point (graded)

Solve $\begin{pmatrix} 1 & -2 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix} = \begin{pmatrix} -4 \\ -3 \\ 2 \end{pmatrix}$

-1 Answer: -1

2 Answer: 2

1 Answer: 1

Solution:

For a complete solution see the notes .

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Homework 1.1.1.3

1/1 point (graded)

In Matlab's command window, create a random upper triangular matrix U:

```
format long

rng( 0 );
n = 3
U = triu( rand( n,n ) )
x = rand( n,1 )
```

Report results in long format. Seed the random number generator so that we all create the same random matrix U and vector x.

```
x = rand( n,1 )
```

```
b = U * x;
```

Compute right-hand side b from known solution x.

```
xhat = U \ b;
```

Solve $U\widehat{x} = b$.

```
xhat - x
```

Report the difference between $\widehat{x}$ and x.

What do we notice? Next, check how close $U\widehat{x}$ is to $b = Ux$:

```
b - U * xhat
```

This is known as the residual. What do we notice?

☑ done

✔

Solution:

For a complete solution see the notes .

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Homework 1.1.1.4

1/1 point (graded)

In Matlab's command window, create a random upper triangular matrix U:

```
format long

rng( 0 );
n = 100;
U = triu( rand( n,n ) );
x = rand( n,1 );
```

Report results in long format. Seed the random number generator so that we all create the same random matrix U and vector x.

```
b = U * x;
```

Compute right-hand side b from known solution x.

```
xhat = U \ b;
```

Solve $U\widehat{x} = b$.

```
norm( xhat - x )
```

Report the Euclidean length of the difference between $\widehat{x}$ and x.

What do we notice? Next, check how close $U\widehat{x}$ is to $b = Ux$, again using the Euclidean length:

```
norm( b - U * xhat )
```

What do we notice?

☑ done

✔

Solution:

For a complete solution see the notes .

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Answers are displayed within the problem

Homework 1.1.1.5

1/1 point (graded)

Continuing with the `U`, `x`, `b`, and `xhat` from the last homework, consider

- When is an upper triangular matrix singular?
- How large is the smallest element on the diagonal of the U from the last homework? (min(abs(diag(U))) returns it!)
- If U were singular, how many solutions to $U\widehat{x} = b$ would there be? How can we characterize them?
- What is the relationship between $\widehat{x} - x$ and U?

What have we learned?

- [x] done

Solution:

For a complete solution see the notes .

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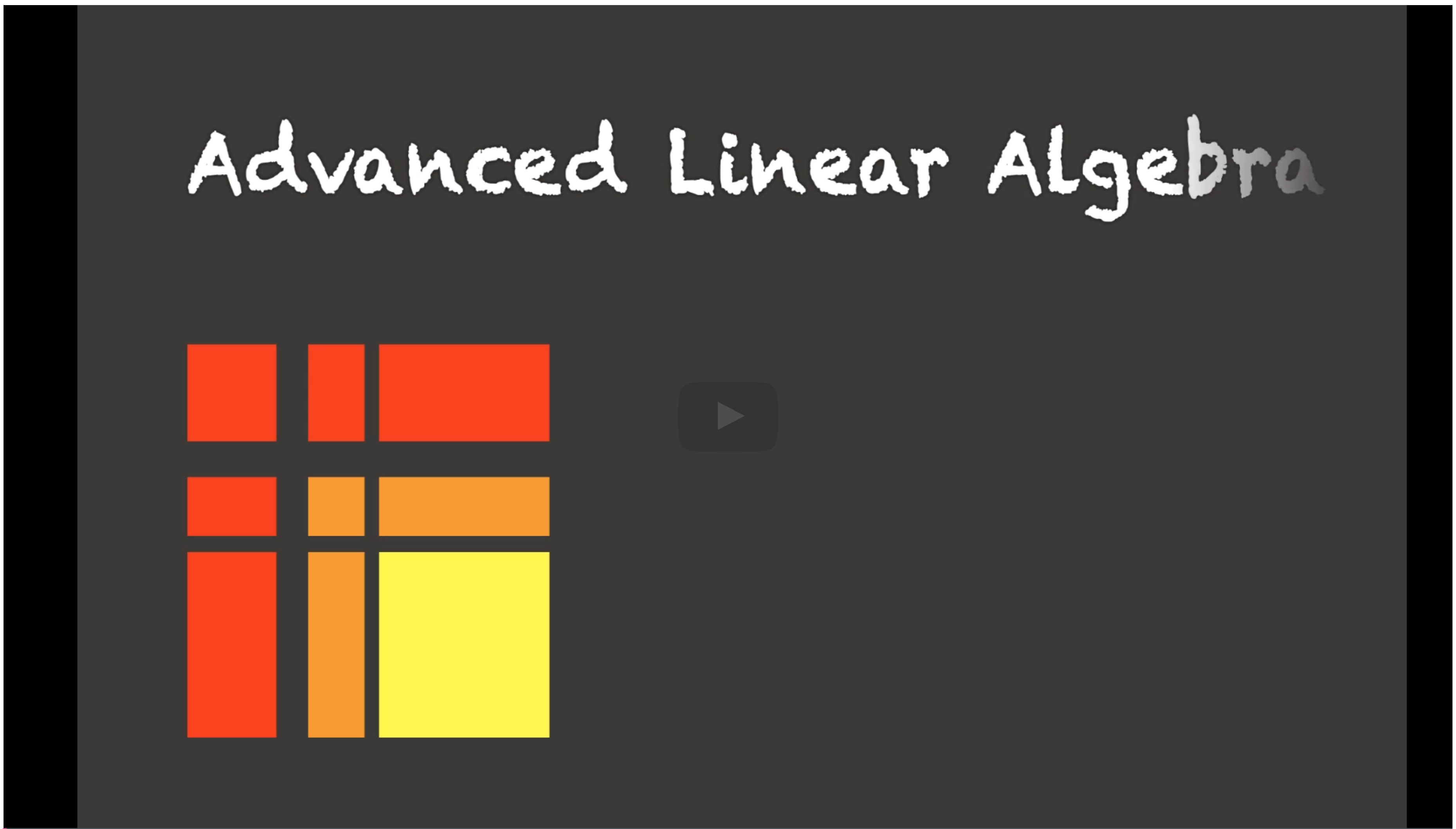

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